

吴筱琦

+86 13776020628 yukiwxq@gmail.com

现居城市：伦敦/苏州

意向城市：上海/苏州/杭州 | 数据分析与生态环境科学



个人总结

帝国理工学院计算生命科学研究型硕士在读生，具备扎实的生命科学背景，系统掌握 R、Python、统计建模、生物信息学及基于大语言模型的数据提取流程。独立设计并实现 VecTraits Curation 项目——一套从科学文献中提取结构化性状数据的端到端 LLM 流程；曾在上海宣泰医药担任分析工程师，主导 HPLC/UV 稳定性检测与质量控制，积累医药研发一线经验。在毕业设计及 11 项独立研究项目中，展现出严谨的定量分析能力与出色的科研写作、跨团队协作能力。希望将统计编程与生物学洞见相结合，为医药研发或临床数据岗位提供数据驱动的分析支持。

教育背景

帝国理工学院 (QS 2) 计算生命科学研究型硕士 伦敦 2025.09 - 2026.09

- **课程内容：**R 语言统计学、生态与进化数据科学、生物计算训练营等生态与进化定量方法核心课程，涵盖统计编程、数据科学、GIS、基因组学与生物信息学
- **独立研究：**开展为期 8 个月的独立研究项目，聚焦编程、建模与统计分析在生态与进化问题中的应用

帝国理工学院 (QS 2) 生物科学本科 伦敦 2022.09 - 2025.07

- **相关课程：**进化与多样性，生物化学与微生物学，生态学与进化论，细胞生物学与遗传学，应用分子生物学，遗传学，生态学基础，行为生态学，生态学实地技能，生物信息学，编程与统计，非洲生物学野外课程，生物多样性与保护生物学，全球变化生物学，法语

实习与社会实践

上海宣泰医药科技有限公司 分析工程师 · 分析部 上海 2024.07 - 2024.09

- **稳定性检测：**使用高效液相色谱-紫外检测 (HPLC/UV) 技术对客户提供的药品样本进行稳定性检测，专注于样品中杂质的检测与分析
- **数据分析：**独立完成样本前处理、HPLC 系统操作及数据分析，确保检测结果的准确性和可靠性
- **质量控制：**通过严格的质控流程鉴别潜在杂质，为药品的稳定性评估提供关键数据支持

上海剑藤教育科技有限公司 教学助教 · 生化部 上海 2023.07 - 2023.08

- **教学辅导：**为冲刺牛津、剑桥等名校的学生提供课外拓展教学辅导，受到国际高中孩子们的欢迎
- **教学执行：**利用本科专业知识，承担课程的部分全英文讲授、批改作业、作业讲解、模拟面试、组织课堂讨论等教学工作，保证教学质量
- **教学成果：**整理和分析输出需求文档，帮助老师根据学生的特点，差生达到中等成绩、优生进一步巩固基础，最终帮助多名学生拿到了剑桥本科录取通知

科研与项目经历

多模态大语言模型提取媒介性状数据评估：高精确率与结构依赖型记录召回 伦敦 2025.12 - 2026.08

Python、Gemma 4 31B IT (多模态 LLM)、PyMuPDF、Docling、提示工程、JSON Schema 校验、二分图记录匹配评估、pytest

- **基准构建：**从 VecTraits 平台超 4 万行、334 篇论文的原始导出数据中，设计加权多样性贪心算法选取 20 篇论文（覆盖不同性状类型、报告格式与数据来源），构建含 2,335 条人工基准记录的开发集与 431 条记录的留出测试集；参照 MIREVT 标准将 91 字段的 VecTraits 原始 schema 精简为 22 字段提取 schema
- **多模态提取流程：**基于 PyMuPDF 提取全文文本、Docling 解析表格结构、150 DPI 渲染图表证据构建多模态证据包，输入 Gemma 4 31B IT (256K 上下文多模态大语言模型)，通过提示工程与 JSON Schema 约束实现从 PDF 文本/表格/图像中可溯源提取原始性状记录；经 11 轮提示词迭代优化，最终版本在开发集上取得 0.810 的条件宏观字段 F1
- **严谨评估与结果：**设计基于二分图记录匹配的评估框架（行集合/条件字段/端到端字段三类指标）及错误溯源分析；在留出测试集上评估锁定流程，取得记录精确率 1.000、召回率 0.360、条件宏观字段 F1 0.729，验证流程“高精确率、结构依赖型召回”特性，论证其适合作为人机协同数据整理的候选记录生成工具

气候变化对佛罗里达州埃及伊蚊种群动态的影响建模 伦敦 2025.03 - 2025.06

R、deSolve、ggplot2、ODEs、统计验证 (RMSE、Pearson、Spearman)

- **模型适配**: 将已发表的热带 ODE 房室模型适配至亚热带地理环境, 模拟佛罗里达 Collier 县埃及伊蚊种群在温度与降雨影响下的动态变化
- **模型验证**: 基于 53 周本地蚊虫监测数据验证模型 (RMSE=1.65, Pearson $r=0.29$, Spearman $\rho=0.30$), 优化时间滞后 ($\tau = -2$ 周) 与降雨尺度因子, 量化模型性能
- **研究成果**: 完成毕业设计, 提出模型改进方向 (引入人为水源、空间异质性与不确定性量化), 为媒介控制与疾病防控研究提供参考

北大西洋渔业数据变化驱动因素分析与食物网构建 伦敦 2025.01 - 2025.02

Excel、文献综述、食物网分析

- **数据分析**: 基于 ICES 北海 8 种鱼类 90 年长期数据, 量化种群动态与物种脆弱性 (如鲽形目鱼类因兼捕与晚熟因素在 1990-2000 年间下降约 40%)
- **文献综合**: 结合文献证据解析政策、气候与渔业活动的影响, 构建食物网并注释生态过程
- **研究成果**: 提出普通海鸦作为指示物种, 报告获高分与优秀评价, 为渔业资源管理提供科学依据

蜂鸟宏观生态学分析: 纬度梯度与生态学规则验证 伦敦 2024.10 - 2024.12

R、ggplot2、caper、GLM、PGLS

- **假设检验**: 基于 AVONET 数据与系统发育树, 检验纬度多样性梯度 (LDG)、伯格曼法则和拉波波特法则
- **建模分析**: 构建物种丰富度热点地图, 并应用 GLM 与 PGLS 分析体重、分布范围与纬度关系
- **研究发现**: 结果强力支持 LDG; 伯格曼法则在普通 GLM 中显著, 但在控制系统发育信号 (PGLS) 后不再显著, 揭示迁徙与生态特化的重要作用

环境花色比例对昆虫颜色选择的影响 开普敦 2024.09 - 2024.10

野外实验、R (t 检验、Pearson 相关)

- **实验设计**: 在南非野外课程中, 于 6 个样地设计五色陷阱盘实验, 研究昆虫访花颜色偏好与环境花色比例的关系
- **实验结果**: 黄色、蓝色和白色陷阱均显著高于基线吸引昆虫 ($p<0.001$); 黄色捕获率与环境黄花比例显著正相关 ($r=0.369$, $p=0.027$)
- **研究结论**: 支持昆虫利用环境中最常见花色作为觅食线索的假说, 深化对授粉生态学的理解

昆虫衰退的驱动因素、影响与保护对策综述 伦敦 2024.03 - 2025.06

文献综述、批判性分析

- **文献综述**: 审阅 70+ 篇文献, 系统分析栖息地丧失、农药使用、气候变化等驱动因素, 其中一项全球荟萃分析显示昆虫生物量每年下降约 2.5%
- **影响评估**: 评估昆虫衰退对生态系统服务 (传粉、养分循环)、农业和人类健康的影响
- **批判分析**: 批判性评估“有得有失”的反叙事及地理/分类学报告偏差, 提出未来研究与政策方向

温度与基质对海葵胆怯行为的影响 普利茅斯 2024.05 - 2024.06

R、Wilcoxon 检验、行为实验

- **实验设计**: 控制温度与基质条件, 测定 52 只海葵的惊愕反应时间 (SRT), 并进行观察者间信度检验 (ICC=0.949)
- **实验结果**: 高温显著缩短 SRT (Wilcoxon 检验 $p=0.017$), 表明更“大胆”的行为表现; 基质无显著效应
- **研究结论**: 研究揭示了环境温度升高与代谢率变化对海洋动物行为的影响

温度暴露时间对蟋蟀攻击行为的影响 伦敦 2024.03 - 2024.03

R、GLM (泊松回归)、行为观察

- **实验设计**: 独立设计实验, 测试不同温度暴露时长对地中海田间蟋蟀攻击行为的影响
- **实验结果**: 高温显著降低遭遇次数 ($p=0.040$) 和持续时间 ($p=0.013$), 暴露时间呈负效应趋势
- **研究结论**: 研究为理解气候变暖对昆虫行为的即时效应提供证据

生态建模与种群动力学模拟 伦敦 2024.01 - 2024.02

R、ggplot2、Lotka-Volterra、逻辑斯蒂增长模型

- **Levins 集合种群模型**: 构建并模拟无限斑块下的种群占领动态, 分析 colonization 与 extirpation 的平衡点
- **渔业捕捞模型**: 基于逻辑斯蒂增长模型, 模拟固定配额与固定努力捕捞策略, 计算最大可持续产量 (MSY)
- **资源竞争模型**: 模拟两种物种对同一资源的竞争过程, 验证 R^* 理论, 分析竞争排斥现象

- **Lotka-Volterra 捕食-被捕食模型**: 将两物种模型扩展为三物种捕食链模型, 引入随机性与时间滞后, 可视化种群周期波动
- **研究成果**: 四个建模项目均获得 90+ 评分 (最高 94 分), 展现扎实的微分方程建模、算法实现与结果可视化能力, 熟练使用 R 进行动态系统模拟、参数优化与多场景分析

近交系小鼠在医学研究中的应用与局限性综述

伦敦 2023.11 - 2023.12

文献综述、遗传学分析

- **文献综述**: 系统阐述近交小鼠的生成原理、医学应用价值与典型研究案例
- **批判分析**: 批判性分析局限性, 包括与人类差异、近交衰退及伦理问题
- **研究成果**: 完成综述论文并获优秀评价, 展示批判性思维与科学写作能力

病毒指纹图谱: 利用 HERV-K solo-LTR 多态性进行人群遗传学分析

伦敦 2023.11 - 2023.12

PCR、凝胶电泳、群体遗传学分析

- **实验操作**: 运用 PCR 和凝胶电泳检测 5 组引物对应的基因组多态性, 结合 F_{ST} 和 HWE (χ^2 检验) 进行群体遗传学分析
- **分析评估**: 评估 HERV-K solo-LTR 作为人群遗传与法医学标记的潜力与挑战
- **研究成果**: 完成实验与数据分析, 成果获优秀评价

OCA1A 型白化病的生物信息学分析

伦敦 2023.10 - 2023.11

NCBI、OMIM、Ensembl、UniProt、AlphaFold、ClinVar

- **结构分析**: 分析 TYR 基因突变对酪氨酸酶结构与功能的影响, 聚焦铜结合域突变, 并使用 AlphaFold 结构预测辅助分析
- **数据库整合**: 使用多数据库进行变异注释、人群频率分析与跨物种比对
- **研究成果**: 完成生物信息学报告, 掌握多数据库整合、蛋白结构预测与进化保守性分析技能

技能/其他

- **Python 与开发工具**: Python (主力语言), Git, 虚拟环境, Claude Code / Cursor 辅助开发; 多后端 LLM 集成 (Gemini API、Ollama 本地推理、OpenAI 兼容接口、HuggingFace 多模态模型)、提示工程、JSON Schema 校验、pytest 测试、基于 JSON 配置/状态跟踪的异步子进程编排、基于环境变量的密钥管理
- **Web 开发**: FastAPI (路由/服务架构)、React、TypeScript、Vite、Tailwind CSS
- **分子与生化实验**: PCR 与凝胶电泳 (多引物基因分型, 如 5 组引物的 solo-LTR 检测)、DNA/RNA 提取 (Qiagen 试剂盒)、蛋白质电泳、西方印迹; 群体遗传学分析 (F_{ST} 、Hardy-Weinberg 平衡/ χ^2 检验); 跨 NCBI、OMIM、Ensembl、UniProt、ClinVar、AlphaFold 的变异注释与蛋白结构分析
- **生态与野外研究**: 野外实验设计 (如多样地彩色陷阱盘偏好实验)、具观察者间信度检验的行为学测定 (行为学评分、惊愕反应时间测量)、生物多样性/环境调查、宏观生态学与系统发育比较分析 (AVONET 级性状数据、PGLS)
- **数据分析与建模**: R (ggplot2、dplyr、caper、ape、vegan、deSolve)、统计推断 (GLM/泊松回归、PGLS、Wilcoxon 检验、t 检验、Pearson/Spearman 相关)、基于微分方程与差分方程的种群动力学建模 (集合种群、捕食-被捕食、逻辑斯蒂增长与 MSY、资源竞争)
- **语言**: 英语 (雅思总分 7.5), 普通话 (母语), 法语 (A2); **兴趣爱好**: 钢琴, 摄影, 滑雪

Xiaoqi Wu

+86 13776020628 yukiwxq@gmail.com

Currently based in: London / Suzhou

Data Analysis & Ecological Sciences | Shanghai / Suzhou / Hangzhou

Summary

MSc Computational Methods in Ecology and Evolution student (Imperial College London) with a strong life-sciences foundation and hands-on skills in R, Python, statistical modelling, bioinformatics, and LLM-based data extraction pipelines. Built an end-to-end LLM pipeline (VecTraits Curation project) for extracting structured trait data from scientific literature, and gained pharmaceutical R&D exposure as an Analytical Engineer at Shanghai Sunherb Pharmaceutical Technology Co., Ltd. (HPLC/UV stability testing, quality control). Combines rigorous quantitative analysis with clear scientific writing and cross-team collaboration, demonstrated across a master's research project and eleven independent research projects. Looking to bring data-driven analytical rigor to a biopharma R&D or clinical data role.

Education

Imperial College London MSc Computational Methods in Ecology and Evolution London 2025.09 - 2026.09

- **Curriculum:** Core quantitative modules in ecology and evolution including Statistics in R, Ecological and Evolutionary Data Science, and Biological Computing Bootcamp, covering statistical programming, data science, GIS, genomics, and bioinformatics
- **Independent Research:** Undertaking an 8-month independent research project on programming, modelling, and statistical analysis applied to ecological and evolutionary questions

Imperial College London BSc Biological Sciences London 2022.09 - 2025.07

- **Relevant Coursework:** Evolution and Diversity, Biochemistry and Microbiology, Ecology and Evolution, Cell Biology and Genetics, Applied Molecular Biology, Genetics, Fundamentals of Ecology, Behavioural Ecology, Field Skills in Ecology, Bioinformatics, Programming and Statistics, Field Course in African Biology, Biodiversity and Conservation Biology, Global Change Biology, French

Internship & Engagement

Shanghai Sunherb Pharmaceutical Technology Co., Ltd. Analytical Engineer · Analytical Department Shanghai 2024.07 - 2024.09

- **Stability Testing:** Conducted stability testing of client-supplied drug samples using HPLC/UV, focusing on impurity detection and analysis
- **Data Analysis:** Independently carried out sample pre-treatment, HPLC system operation, and data analysis, ensuring accuracy and reliability of test results
- **Quality Control:** Identified potential impurities through rigorous quality-control procedures, providing key data to support drug stability assessments

Shanghai Jianteng Education Technology Co., Ltd. Teaching Assistant · Biochemistry Department Shanghai 2023.07 - 2023.08

- **Tutoring:** Provided extracurricular tutoring for students preparing for Oxford, Cambridge, and other top universities, well received by international high-school students
- **Teaching Delivery:** Delivered part of the course fully in English, graded assignments, reviewed homework, conducted mock interviews, and led class discussions, drawing on undergraduate subject knowledge
- **Results:** Compiled and analyzed requirement documents to help teachers tailor instruction to student needs; helped multiple students secure Cambridge undergraduate offers

Research & Project Experience

Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery London 2025.12 - 2026.08

Python, Gemma 4 31B IT (multimodal LLM), PyMuPDF, Docling, prompt engineering, JSON Schema validation, bipartite record-matching evaluation, pytest

- **Benchmark Construction:** Designed a weighted greedy diversity-selection algorithm to select 20 papers (spanning trait types, reporting formats, and data sources) from VecTraits' 40,000+-row, 334-paper raw export; built a 2,335-record manually curated development benchmark and a 431-record held-out test set; reduced the 91-column VecTraits schema to a 22-field extraction schema aligned with the MIREVTD reporting standard
- **Multimodal Extraction Pipeline:** Built a multimodal evidence pipeline combining PyMuPDF full-text extraction, Docling table-structure parsing, and 150 DPI figure/table rendering, fed into Gemma 4 31B IT (a 256K-context multimodal LLM) to extract provenance-linked trait records from PDF text, tables, and images via prompt engineering and JSON Schema-constrained output; iterated across 11 prompt versions, with the final version reaching a development-set conditional macro field F1 of 0.810
- **Rigorous Evaluation & Results:** Designed a bipartite record-matching evaluation framework (row-set, conditional-field, and end-to-end field metrics) with systematic error analysis; evaluated the locked pipeline on the held-out test set, achieving row precision of 1.000, row recall of 0.360, and conditional macro field F1 of 0.729 – confirming a "high-precision, structure-dependent recall" profile and supporting the pipeline's use as a candidate-record generator within a human-in-the-loop curation workflow

Modelling the Impact of Climate Change on *Aedes aegypti* Population Dynamics in Florida

London 2025.03 - 2025.06

R, deSolve, ggplot2, ODEs, statistical validation (RMSE, Pearson, Spearman)

- **Model Adaptation:** Adapted a published tropical ODE compartment model to a subtropical geography to simulate *Aedes aegypti* population dynamics in Collier County, Florida under temperature and rainfall effects
- **Validation:** Validated the model against 53 weeks of local mosquito-surveillance data (RMSE=1.65, Pearson $r=0.29$, Spearman $\rho=0.30$), optimizing time-lag ($\tau = -2$ weeks) and a rainfall scaling factor to quantify model performance
- **Outcome:** Completed final-year project proposing model improvements (incorporating anthropogenic water sources, spatial heterogeneity, and uncertainty quantification) to inform vector control and disease prevention research

Drivers of Change in North Atlantic Fisheries Data and Food Web Construction

London 2025.01 - 2025.02

Excel, literature review, food web analysis

- **Data Analysis:** Analyzed a 90-year time series across 8 fish species from ICES North Sea surveys, quantifying species vulnerability (e.g. sole declined 40% from 1990-2000 due to bycatch and slow maturity)
- **Synthesis:** Synthesized literature evidence on the effects of policy, climate, and fishing activity; constructed and annotated a food web of ecological processes
- **Outcome:** Proposed the common guillemot as an indicator species; report received high marks and distinction, informing fisheries resource management

Macroecology of Hummingbirds: Testing Latitudinal Gradients and Ecological Rules

London 2024.10 - 2024.12

R, ggplot2, caper, GLM, PGLS

- **Hypothesis Testing:** Tested the latitudinal diversity gradient (LDG), Bergmann's rule, and Rapoport's rule using AVONET data and a phylogenetic tree
- **Modelling:** Built species-richness hotspot maps and applied GLM and PGLS to analyze relationships between body mass, range size, and latitude
- **Findings:** Results strongly supported the LDG; showed Bergmann's rule was significant under a naive GLM but lost significance once phylogenetic signal was controlled for (PGLS), highlighting the role of migration and ecological specialization

Effect of Environmental Flower Colour Ratios on Insect Colour Preference

Cape Town 2024.09 - 2024.10

Field experiment, R (t-test, Pearson correlation)

- **Experimental Design:** Designed a 5-colour pan-trap field experiment across 6 sites during a South Africa field course to study insect flower-visiting colour preferences relative to environmental flower colour ratios

- **Results:** Yellow, blue, and white traps all significantly attracted insects over baseline ($p < 0.001$); yellow capture rate was positively correlated with the proportion of yellow flowers in the environment ($r = 0.369$, $p = 0.027$)
- **Conclusion:** Supported the hypothesis that insects use the most common flower colour in the environment as a foraging cue, deepening understanding of pollination ecology

Review: Drivers, Impacts, and Conservation Strategies for Insect Decline London 2024.03 - 2025.06

Literature review, critical analysis

- **Literature Review:** Reviewed 70+ papers, systematically analyzing drivers such as habitat loss, pesticide use, and climate change, including a global meta-analysis reporting 2.5%/year insect biomass decline
- **Impact Assessment:** Assessed the impact of insect decline on ecosystem services (pollination, nutrient cycling), agriculture, and human health
- **Critical Analysis:** Critically evaluated a "winners and losers" counter-narrative and geographic/taxonomic reporting bias, proposing future research and policy directions

Effects of Temperature and Substrate on Sea Anemone Startle Behaviour Plymouth 2024.05 - 2024.06

R, Wilcoxon test, behavioural experiment

- **Experimental Design:** Measured startle response time (SRT) of 52 sea anemones under controlled temperature and substrate conditions, with inter-observer reliability testing ($ICC = 0.949$)
- **Results:** Elevated temperature significantly shortened SRT (Wilcoxon $p = 0.017$), indicating "bolder" behaviour; substrate had no significant effect
- **Conclusion:** Findings reveal the effects of rising environmental temperature and metabolic rate changes on marine animal behaviour

Effect of Temperature Exposure Duration on Cricket Aggressive Behaviour London 2024.03 - 2024.03

R, GLM (Poisson regression), behavioural observation

- **Experimental Design:** Independently designed an experiment to test the effect of different temperature exposure durations on aggressive behaviour in Mediterranean field crickets
- **Results:** High temperature reduced encounter frequency ($p = 0.040$) and duration ($p = 0.013$); exposure time showed a negative-effect trend
- **Conclusion:** Provided evidence for understanding the immediate effects of warming on insect behaviour

Ecological Modelling and Population Dynamics Simulation London 2024.01 - 2024.02

R, ggplot2, Lotka-Volterra, logistic growth model

- **Levins Metapopulation Model:** Built and simulated occupancy dynamics across infinite patches, analyzing the balance between colonization and extirpation
- **Fisheries Harvesting Model:** Simulated fixed-quota and fixed-effort harvesting strategies based on a logistic growth model, calculating maximum sustainable yield (MSY)
- **Resource Competition Model:** Simulated competition between two species for a shared resource, validating R^* theory and analyzing competitive exclusion
- **Lotka-Volterra Predator-Prey Model:** Extended a 2-species model to a 3-species predation chain with stochasticity and time lags, visualizing population cycles
- **Results:** All four modelling projects scored 90+ (up to 94), demonstrating solid skills in differential equation modelling, algorithm implementation, and results visualization; proficient in using R for dynamic system simulation, parameter optimization, and multi-scenario analysis

Review: Applications and Limitations of Inbred Mouse Strains in Medical Research London 2023.11 - 2023.12

Literature review, genetic analysis

- **Literature Synthesis:** Systematically described the generation principles, medical applications, and representative case studies of inbred mice
- **Critical Analysis:** Critically analyzed limitations, including differences from humans, inbreeding depression, and ethical issues
- **Outcome:** Completed review paper with distinction, demonstrating critical thinking and scientific writing skills

Viral Fingerprinting: Population Genetics Analysis Using HERV-K solo-LTR Polymorphism

London 2023.11 - 2023.12

PCR, gel electrophoresis, population genetics analysis

- **Lab Work:** Detected genomic polymorphisms across 5 primer sets using PCR and gel electrophoresis; performed population genetics analysis with F_{ST} and HWE (χ^2 tests)
- **Analysis:** Evaluated the potential and challenges of HERV-K solo-LTR as a population genetic and forensic marker
- **Outcome:** Completed experiments and data analysis; results received distinction

Bioinformatic Analysis of OCA1A Albinism

London 2023.10 - 2023.11

NCBI, OMIM, Ensembl, UniProt, AlphaFold, ClinVar

- **Structural Analysis:** Analyzed the effect of TYR gene mutations on tyrosinase structure and function, focusing on copper-binding domain mutations, using AlphaFold structural predictions
- **Database Integration:** Used multiple databases for variant annotation, population frequency analysis, and cross-species comparison
- **Outcome:** Completed a bioinformatics report, developing skills in multi-database integration, protein structure prediction, and evolutionary conservation analysis

Skills & Other

- **Python & Tooling:** Python (primary language), Git, virtual environments, Claude Code / Cursor for AI-assisted development; multi-backend LLM integration (Gemini API, Ollama local inference, OpenAI-compatible APIs, HuggingFace multimodal models), prompt engineering, JSON Schema validation, pytest testing, async subprocess orchestration with JSON-based config/state tracking, environment-variable-based API key management
- **Web Development:** FastAPI (routers/services architecture), React, TypeScript, Vite, Tailwind CSS
- **Molecular & Biochemical Techniques:** PCR and gel electrophoresis (multi-primer genotyping, e.g. 5-primer-set solo-LTR detection), DNA/RNA extraction (Qiagen kits), protein electrophoresis, Western blotting; population genetics analysis (F_{ST} , Hardy-Weinberg equilibrium/ χ^2 tests); variant annotation & protein structure analysis across NCBI, OMIM, Ensembl, UniProt, ClinVar, and AlphaFold
- **Ecology & Fieldwork:** field experimental design (e.g. multi-site pan-trap colour-preference experiments), behavioural assays with inter-observer reliability testing (ethogram scoring, startle-response measurement), biodiversity/environmental surveying, macroecology and phylogenetic comparative analysis (AVONET-scale trait data, PGLS)
- **Data Analysis & Modelling:** R (ggplot2, dplyr, caper, ape, vegan, deSolve), statistical inference (GLM/Poisson regression, PGLS, Wilcoxon signed-rank, t-tests, Pearson/Spearman correlation), ODE- and difference-equation-based population dynamics modelling (metapopulation, predator-prey, logistic growth & MSY, resource competition)
- **Languages:** English (IELTS 7.5 overall), Mandarin Chinese (native), French (A2) | **Interests:** piano, photography, skiing